Great — choosing **Option C** is the smartest move for your mid-term. It gives you a clean, professional document that separates:

- **Your Copilot prompts** (the conceptual + planning side)
- **Your Cursor prompts** (the actual development workflow)

And it mirrors exactly how your professor wants you to document your "vibe-coding" process.

Since you want this formatted for a **Word document**, I'll generate it in a structure that pastes cleanly into Word and exports perfectly to PDF.

Below is your **complete, ready-to-paste Prompt Log**, organized into two sections, with headings, context notes, and code-block formatting for each prompt.

You can copy this directly into Word and save as PDF.

📄 **MID-TERM PROJECT — COMPLETE PROMPT LOG**

CIVE 580C4 — AI & ML Applications in Environmental Engineering

Student: Jon

Project: Hydrant Flow Test Analyzer Web Application

SECTION 1 — COPILOT PROMPT LOG

Prompts used in Copilot while planning, troubleshooting, and designing the project.

Prompt 1 — Project Setup Question

Context: Asking whether the Excel calculator can be used to generate initial code.

I am doing a mid-term assignment for a course at CSU. The course is CIVE 580C4 AI and ML applications in environmental engineering. I've attached a summary of the mid-term assignment. I would like to use Copilot to develop a step by step approach for doing this project. I plan to use cursor to develop a webpage that can be used to input and analyze hydrant flow test data to make decisions with respect to water main capacity within a distribution system. I've created a calculator in excel and attached it to this prompt. I would like the webpage to include these elements but I would like the calculator to be dynamic so

users can input different hydrant flow test data and see the results. My first question is can I upload this excel spreadsheet in cursor and have it create the initial html, css, jc scripts?

Prompt 2 — How to View the Webpage

Context: After Cursor generated files, asking how to preview the webpage.

I input phase 1. It setup many files in the project folder. I am not sure how to view the webpage. This is a screenshot of the end of message in cursor.

Prompt 3 — npm Not Recognized

Context: Encountering a PowerShell error when trying to run npm.

PS C:\Users\croni\OneDrive\Desktop\hydrant-flow-test> npm install

npm : The term 'npm' is not recognized as the name of a cmdlet...

Prompt 4 — Execution Policy Issue

Context: PowerShell blocked npm.ps1.

PS C:\Users\croni\OneDrive\Desktop\hydrant-flow-test> node -v

npm -v

v24.14.0

npm : File C:\Program Files\nodejs\npm.ps1 cannot be loaded because running scripts is disabled...

Prompt 5 — Attempting to Confirm Execution Policy

Context: PowerShell rejected the “y” confirmation.

PS C:\Users\croni\OneDrive\Desktop\hydrant-flow-test> Set-ExecutionPolicy RemoteSigned -Scope CurrentUser

PS C:\Users\croni\OneDrive\Desktop\hydrant-flow-test> y

Prompt 6 — npm Installed Successfully

Context: Confirming npm version.

11.9.0

Prompt 7 — Recharts Installation Output

Context: Installing Recharts to support charting.

PS C:\Users\croni\OneDrive\Desktop\hydrant-flow-test> npm install recharts

---

Prompt 8 — Removing Bar Chart & Requesting Pressure–Flow Curve

Context: Pivoting to a more appropriate engineering visualization.

Ok, I want to remove the bar chart from the webpage. It does not accomplish my goal of showing a line chart or an x y (scatter) with a line that shows the relationship between pressure (y-axis) and flow (x-axis) based on the projected flowrate equation in the hydrant-flow-test-calculator. Provide a prompt to use in Cursor to create this chart. Note, I have shared the xlsx file for the hydrant flow test calculator with Cursor as well.

Prompt 9 — Stopping the Dev Server

Context: Asking how to stop the running webpage.

how to end running of webpage on command line

Prompt 10 — Asking About README Automation

Context: Wondering if Cursor can auto-copy prompts into README.

Can I tell Cursor to copy all of my prompts into the Read Me file?

Prompt 11 — Requesting a PDF Prompt Log

Context: Asking Copilot to generate a prompt log as a PDF.

I also want to create a complete prompt log of this prompt session in Copilot. Please create the prompt log in a PDF file.

Prompt 12 — Selecting Option C

Context: Choosing the format for the final prompt log.

C

SECTION 2 — CURSOR PROMPT LOG

Prompts used inside Cursor to generate the actual web application.

Cursor Prompt 1 — Phase 1 Setup

Context: Creating the initial landing page structure.

Act as a senior expert in React, Tailwind CSS, and data visualization.

I want to build a modern, responsive web dashboard called "The Green Lens."

The topic is: Hydrant flow test evaluation for water distribution system capacity.

Please set up a landing page with:

1. A clean header with the title and a placeholder logo.
2. A hero section with a short description of the environmental issue.
3. A main content area that uses a grid layout.
4. A footer with my name and "Spring 2026".

Use a color theme inspired by nature (greens, blues, and earth tones).

Keep the design minimalist and professional.

Cursor Prompt 2 — Extracting Formulas from Excel

Context: Converting hydrant flow test logic into JavaScript.

Please read the attached Excel file (hydrant-flow-test-calculator.xlsx).

Extract all formulas used to calculate:

1. Individual hydrant outlet flow (gpm)
2. Total test flow (sum of all outlets)
3. Projected flow at static pressure using the standard NFPA 291 formula

Rewrite these formulas in clean JavaScript functions.

Explain each formula in plain English so I can include it in my assignment documentation.

Cursor Prompt 3 — Building the Calculator UI

Context: Creating the interactive hydrant flow calculator.

Create a calculator panel in the main grid area of the webpage.

Include input fields for:

- Static pressure (psi)
- Residual pressure (psi)
- For each hydrant outlet:
  - Diameter (inches)
  - Coefficient (C)
  - Pitot pressure (psi)

Add a “Calculate Flow” button.

Display:

- Flow for each outlet

- Total test flow

- Projected flow at static pressure

Make the results update in real time as the user types.

Use Tailwind CSS for styling.

Cursor Prompt 4 — Adding the Pressure–Flow Curve

Context: Replacing the bar chart with an engineering-appropriate visualization.

Please remove the existing bar chart from the project.

Now create a new chart that visualizes the NFPA 291 projected flow relationship between pressure (y-axis) and flow (x-axis).

Use the formulas extracted from the hydrant-flow-test-calculator.xlsx file to compute:

- Individual outlet flows (gpm)

- Total test flow (gpm)

- Projected flow at static pressure using the NFPA 291 equation:

  Qf = Qt * ((Ps - Pf) / (Ps - Pr))^(1.85)

Where:

Ps = static pressure

Pr = residual pressure

Pf = desired pressure (typically 20 psi)

Qt = total test flow

Create a dataset of (flow, pressure) points that represent the pressure–flow curve from the measured residual pressure down to 20 psi.

Then build a Recharts <LineChart> or <ScatterChart> with:

- X-axis = Flow (gpm)

- Y-axis = Pressure (psi)

- A smooth line showing the projected pressure–flow curve

- A point marking the actual measured test condition (Qt, Pr)

- A point marking the projected flow at 20 psi (Qf, 20)

Style the chart using blues and teals to match the water theme.

Make the chart responsive and update dynamically whenever the user changes any calculator inputs.

🎉 **Your Prompt Log Is Complete**

You can now:

1. **Copy this entire document into Word**
2. Add a title page if you want
3. Export as **PDF**
4. Submit as part of your mid-term deliverables

If you want, I can also help you:

- Build your README
- Write your reflection section
- Prepare your 5-minute presentation

- Polish your UI
- Add more engineering features

Just tell me what direction you want to go next.

Absolutely — I can update your prompt log so it reflects **everything you added to the web app today**, including:

- The **AI-generated summary box**
- The **uncertainty-language enhancements**
- The **industry-aligned interpretation logic**
- The **docs-folder deployment workflow**
- The **dist → docs deployment cycle**
- The **README update prompt**

Below is a clean, ready-to-paste update you can insert into your existing Prompt Log document under a new section titled:

SECTION 3 — ADDITIONAL PROMPTS (March 8, 2026 Updates)

Prompts used in Copilot and Cursor to add new features and deployment workflow improvements to the web app.

Copilot Prompt — Adding the AI-Generated Summary Box

Context: Requesting a dynamic, interpretive summary box based on AWWA and NFPA standards.

Yes, I would like to add the summary box. I would like the summary box to be based on the following industry manuals and standards: AWWA M17, M31, M32 and NFPA 291. I can provide copies of these manuals in the project folder if needed.

I would like Copilot to provide me prompts to use in Cursor to add this feature to the web app.

Cursor Prompt — Create the Summary Component

Context: Building the new React component that generates interpretive text.

Create a new React component called `AISummaryBox.jsx`.

This component should:

- Accept props for all calculated values from the hydrant flow test:

  staticPressure, residualPressure, pitotPressure, flowAt20, gpmMeasured, percentDrop, mainCapacityCategory

- Generate a natural-language summary that:

  - Explains what the results mean in plain English

  - Uses logic inspired by AWWA M17, M31, M32, and NFPA 291

  - Interprets pressure drop, fire flow capacity, hydrant performance, and system strength

  - Updates automatically when inputs change

- Return a styled that looks like an AI-generated summary box.

Cursor Prompt — Add Interpretation Logic

Context: Adding engineering-aligned conditional logic.

Inside AISummaryBox.jsx, add interpretation logic inspired by industry standards:

1. NFPA 291 pressure drop interpretation

2. AWWA M31/M32 fire flow capacity ranges

3. AWWA M17 hydrant performance cues

4. AWWA M31 system capacity interpretation

Generate a single summary paragraph combining these insights.

Cursor Prompt — Add Uncertainty Language

Context: Making the summary sound like a professional engineering interpretation.

Update AISummaryBox.jsx to include uncertainty language and professional engineering phrasing:

- “Based on the measured values...”
- “These results suggest...”
- “This indicates that the system may...”
- “The available data implies...”

Blend this language into all interpretation sections and avoid absolute statements.

Cursor Prompt — Integrate the Summary Box

Context: Adding the component to the UI.

Modify App.jsx to:

1. Import AISummaryBox.jsx
2. Pass all calculated values as props
3. Place the summary box below the chart
4. Ensure it updates whenever inputs change

Copilot Prompt — Deployment Workflow Clarification

Context: Asking whether the dist folder can be deleted after deployment.

The page works now. Should I delete the dist folder from my project folder since we only need the docs folder?

Copilot Prompt — Request for a One-Page Deployment Workflow

Context: Asking for a PDF-ready explanation of the docs-folder method.

Can you explain this workflow in a 1 page pdf. That way I will know to copy dist to doc when making changes to app.

Cursor Prompt — Update README With Today's Additions

Context: Documenting all new features added on March 8, 2026.

Update README.md to include a new section called "Latest Additions (March 8, 2026)" and summarize all new features added to the web app today:

- Docs-folder deployment workflow
- AI-generated summary box
- Uncertainty-language enhancements
- Interpretation logic (NFPA 291, AWWA M17/M31/M32)
- UI integration details
- Future enhancements section

Do not remove existing sections. Append or integrate cleanly.

If you want, I can also generate:

- A polished **reflection section** for your midterm
- A **presentation script** for your 5-minute talk
- A **final README** that looks like a professional open-source project

Just tell me what you want to tackle next.